Huawei Access Node KR-ACC-72 OSPF Routing Protocol Flap Troubleshooting SOP

Node: KR-ACC-72 | **Vendor:** Huawei | **Severity:** Minor | **Alarm Code:** ALARM_3936

Description

This SOP describes the procedures to troubleshoot OSPF routing protocol flap failures on Huawei Access Node KR-ACC-72 at KR Puram Access site. OSPF (Open Shortest Path First) is an IGP (Interior Gateway Protocol) used for routing within an autonomous system. The OSPF protocol flap is characterized by a series of rapid OSPF adjacency losses and reconstructions, which in turn cause BGP (Border Gateway Protocol) route churn, leading to suboptimal routing and potential network instability. The OSPF protocol flap on Huawei Access Node KR-ACC-72 can be caused by various factors, including but not limited to, hardware or software errors, configuration inconsistencies, or environmental factors such as high temperature or humidity. In this SOP, we will cover the diagnostic procedures to identify the root cause of OSPF protocol flap and provide step-by-step instructions to resolve the issue.

Pre-requisite Checks

- Check the temperature and humidity levels at the KR Puram Access site to ensure they are within the recommended operating range for Huawei Access Node KR-ACC-72.

- Verify the power supply to the node and ensure that the redundant power module is functioning correctly.

- Check the link status of the OSPF-enabled interfaces and ensure that there are no link flaps or errors.

Potential Root Causes

Cause	Probability
Hardware or software errors	High
Configuration inconsistencies	Medium
Environmental factors	Low

Diagnostic Commands & Expected Outputs

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Command: display ospf peer, Expected Output: OSPF peer information, Description: This command displays the OSPF peer information, including the peer ID, state, and authentication information. It helps to identify any OSPF adjacency losses or reconstructions and provides insights into the OSPF protocol flap.

Command: display ospf neighbor, Expected Output: OSPF neighbor information, Description: This command displays the OSPF neighbor information, including the neighbor ID, state, and authentication information. It helps to identify any OSPF adjacency losses or
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reconstructions and provides insights into the OSPF protocol flap.

Command: display bgp routing-table, Expected Output: BGP routing table information, Description: This command displays the BGP routing table information, including the prefix, next hop, and AS path information. It helps to identify any BGP route churn caused by the OSPF protocol flap.
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Resolution Steps

Step 1: Step: Check OSPF configuration, Command: display ospf configuration, Description: This command displays the OSPF configuration information, including the OSPF area, router ID, and network information. It helps to identify any configuration inconsistencies that may be causing the OSPF protocol flap.

Step 2: Step: Check OSPF interface status, Command: display ospf interface, Description: This command displays the OSPF interface status information, including the interface ID, state, and authentication information. It helps to identify any OSPF adjacency losses or reconstructions and provides insights into the OSPF protocol flap.

Step 3: Step: Reset OSPF process, Command: reset ospf, Description: This command resets the OSPF process, which helps to resolve the OSPF protocol flap by clearing the OSPF adjacency losses and reconstructions.

Post-Check Verification

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Command: display ospf peer, Expected Output: OSPF peer information, Description: This command displays the OSPF peer information, including the peer ID, state, and authentication information. It helps to verify that the OSPF protocol flap has been resolved and that the OSPF adjacency is stable.

Command: display bgp routing-table, Expected Output: BGP routing table information, Description: This command displays the BGP routing table information, including the prefix, next hop, and AS path information. It helps to verify that the BGP route churn has been resolved and that the routing is stable.
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Roll-back Procedures

- Step: Restore OSPF configuration, Command: restore ospf configuration, Description: This command restores the original OSPF configuration, which helps to roll back the changes made to resolve the OSPF protocol flap.

- Step: Reset OSPF process, Command: reset ospf, Description: This command resets the OSPF process, which helps to roll back the changes made to resolve the OSPF protocol flap.